

Decoding Books Reviews: Analyzing Features of Popular Books, A Specific Genre of Books, and Reader Sentiment

Text Mining Final Project – Using Amazon Books Reviews Dataset

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ABSTRACT

In the project, we attempt to explore the application of text mining and natural language processing (NLP) techniques to analyze user-generated content in the form of Amazon Books Reviews. The primary objective is to discover the keywords and sentiments expressed by users, thereby gaining insights into certain aspects of book reception and preferences. Specifically, the study aims to answer several critical questions concerning the characteristics of popular books, the categorization of certain types of books, and the common grievances aired by customers. By extracting and analyzing keywords and sentiment data from reviews, this research will provide a preliminary overview of consumer behaviors and preferences, potentially guiding publishers and authors in tailoring their offerings to meet market demands more effectively. Through a systematic examination of textual data from reviews, this project will contribute to the broader understanding of contemporary reading habits and publishing trends.

KEYWORDS

Amazon Books Reviews, Natural Language Processing, Keywords Extraction, Sentiment Analysis, TF-IDF

1 Introduction

1.1 About Dataset

In our project, we used a dataset from Kaggle, a popular platform that provides a community where users can collaborate on projects, share code, and access various datasets. The specific dataset we are working with is the "Amazon Books Reviews" dataset, which consists of two CSV files named "Books_rating.csv" and "books_data.csv". These files include title, description, authors, image, preview link, publisher, published date, categories, ratings count of 212404 unique books and at the same time they have 3 million reviews of users for them including rating scores, review score, review time,... With the "Amazon Books Reviews" dataset from Kaggle, we have a rich source of information to explore and gain insights into the world of books and their reviews. By applying our knowledge of Text Mining techniques,

we hope to uncover valuable patterns and trends that can help us better understand this vast and diverse dataset.

1.2 Motivation

Every year, countless books are published, ranging from hundreds of thousands to millions. However, only a select few achieve popularity and recognition among readers. This raises an important question within the publishing industry: what factors, aside from content, contribute to a book's success? In other words, what motivates individuals to invest their money and time in reading and reviewing a particular book?

To gain deeper insights into the genres of books that sell well on Amazon, one can explore the Amazon Books Reviews dataset. This dataset provides reliable and quantitative answers to questions about book trends and customer behavior over the years. However, handling the Amazon Books Reviews dataset, like any other dataset, presents its own challenges. Dealing with large amounts of missing data and inefficient data naming are two significant hurdles. Additionally, grappling with unstructured data poses an additional challenge. While the majority of comments are in English, there is also a small fraction of documents in Spanish or French within this extensive dataset. Therefore, it is important to consider these non-English entries, and using text mining techniques becomes necessary in order to categorize and analyze the sentiments expressed by readers and reviewers. Furthermore, during the exploration of the most popular book categories based on the number of books classified under a specific genre, a peculiar category named "Books" was discovered. Surprisingly, this category contains over 500 books. This discovery raises the question of what these books actually include and how they can be described based on review content, particularly given that a significant portion lacks descriptions or any columns indicating their primary content. By exploring customers who gave a score of 1 for a popular book in the "Books" category, titled "The Death of Outrage: Bill Clinton and the Assault on American Ideals," we also found that the number of people who scored this book as 1/5 is in the second position, just after those who scored it 5/5. This is an unusual situation and quite different from the general trend, where the number of 4/5 scores is typically in the second position after 5/5 scores, not 1/5.

This makes us even more curious about what might have influenced the customers to feel this way.

1.3 Questions

The following questions were derived from the problem described in motivation part:

Q1: What are features of popular books?

Q2: What kinds of books are categorized as ["Books"]? Can we use text mining techniques to classify them into meaningful single genre?

Q3: What are customer complaining about books in review?

1.4 Solution – Text Mining Approach

Text mining is the process and technique used for searching, retrieving, and extracting high-quality, useful, and purposeful information from the ocean of unstructured and unclassified data and information in the form of text written in natural language [1]. Understanding this powerful function of text mining, we believe that applying its techniques to the Amazon Books Reviews dataset is necessary work in order to find answers to our questions and gain more insights from this big e-commerce platform. Compared to the past, there are now more and more programming languages and mining techniques that can support us in many domains to slice and dice the data that we have. You can use any language that you are familiar with, such as R, Python, Java, SQL, with their extensive and vast libraries, frameworks, and packages such as quanteda, text2vec, NLTK, gensim, scikit-learn, depending on your specific goals in exploring the data. Our aim is to answer three questions that we are curious about in the problem statement by implementing those text mining techniques. Among the many programming languages out there, we have decided to utilize Python as the main programming language because of its simplicity and effectiveness, especially with its suitability for linguists and natural language toolkits, which will give us a significant advantage in gaining more knowledge from the dataset or in exploring ideas for future projects.

2 Literature Reviews

In order to provide a solid theoretical foundation for this project and assess the current state of the art, we have consulted various academic research papers on text mining, web mining in the library domain, and other relevant fields. The work by Antons et al. [2] and Udgave et al. [3] are notable examples that highlight the value of NLP techniques for text processing, demonstrating their impressive impact in extracting insights from unstructured data and showcasing the evolution of text mining research in innovative studies. Particularly, the paper by Liu et al. [4] not only explores the public's perception of educational books through text mining reviews on Douban Books, but also employs keyword analysis, topic analysis, and aspect-level sentiment analysis. This paper provides valuable insights into various aspects of the education industry, emphasizing the importance of identifying user attitudes and utilizing online comments for text analysis. It has greatly contributed to enriching and enhancing our

frameworks, aiding in the development of a more streamlined flowchart for handling large volumes of data, and providing a deeper understanding of the current advantages and challenges associated with text mining techniques.

3 Methods

3.1 Text Preprocessing

Text preprocessing is an initial step in natural language processing (NLP) that includes cleaning and transforming unstructured text data to prepare it for analysis. In text preprocessing steps we have to do many tasks which aim to make our data cleaner and easier for future analyzing and these steps contains noise reduction, normalization, tokenization, stopwords, lemmatization such. In noise reduction step, we need to get rid of punctuation, special characters or symbols, deleting unnecessary information like url,...because it will like to other next step when machine will treat every word (token) equally and noise reduction can avoid unexpected result. Tokenization process will be break text data into smaller units, this helps machine understand human language and enabling feature extraction. Tokenization also have different types like Word tokenization, character tokenization, subword tokenization. In the next step, we also need to remove words which are often occur in the documents but carry little meaning like "a", "an", "the", "is", "or" and "and". Last but not least in the project, we also reduce words to their base or root form as stemming and lemmatization, while stemming is a techniques to remove suffixes of a word, lemmatization is more sophisticated processing as using vocabulary and morphological analysis to determine that word's base form.

3.2 Exploratory Data Analysis (EDA)

After executing many step of preprocessing our documents/corpus, we describe the data by means of statistical visualization techniques by EDA because firstly we try to avoid assumption before analyzing and secondly we want to have a general look of data to decide which part of dataset that we will focus on. We choose to use charts and graphs to spot trends and displaying information and relationship between variables in data.

3.3 Keywords Extraction

3.3.1 TF-IDF

Besides rigorous approaches to natural language processing (NLP) that included such things as sentence segmentation, tokenization, lemmatization, word chunking, and entity detection are necessary in order to achieve the deepest possible understanding of textual data, it is also important to imply techniques from Information Retrieval theory such as TF-IDF. TF-IDF stands for term frequency-inverse document frequency and can be used to query a corpus by calculating normalized scores that express the relative important of terms in the document [5]. We utilize TF-IDF in this project to extract important words among reviews of Amazon customers.

3.3.2 Topic Modeling using *Latent Dirichlet Allocation (LDA)*

Topic modeling is an unsupervised machine learning technique that's capable of scanning a set of documents, detecting word and phrase patterns within them, and automatically clustering word groups and similar expressions that best characterize a set of documents.[6] There have many methods of topic modeling such as Latent Dirichlet Allocation (LDA), Latent Semantic Allocation (LSA) or Non-negative Matrix-Factorization (NNMF). Latent Dirichlet Allocation is the only topic modeling method that we use in this project, that classifies or categorizes the text into a document and the words per topic, these are modeled based on the Dirichlet distributions and processes. The LDA makes two key assumptions: Documents are a mixture of topics and Topics are mixture of tokens (or words)[7] and the generative process for LDA corresponds to the following joint distribution of the hidden and observed variables[8]

$$p(\beta_{v,K}, \theta_{v,D}, z_{v,D}, w_{v,D}) = \prod_{i=1}^K p(\beta_i) \prod_{d=1}^D p(\theta_d) \left(\prod_{n=1}^N p(z_{d,n} | \theta_d) p(w_{d,n} | \beta_{v,K}, z_{d,n}) \right).$$

- α (alpha): Proportion parameter.
- η (eta): Topic parameter.
- β_K is a distribution over the vocabulary.
- $\theta_{d,K}$ is the topic proportion for topic k in document d .
- $z_{d,n}$ is the topic assignment for the n -th word in document d
- $w_{d,n}$ is the n -th word in document d , which is an element from the fixed vocabulary.

3.3.3 The coherence score

Coherence score is a statistical measure used to evaluate the quality and interpretability of topics generated by models like Latent Dirichlet Allocation (LDA). It quantifies how semantically related the high-scoring words in a topic are, reflecting how meaningful and coherent these topics are to human readers. The score is calculated by examining the co-occurrence and probability distribution of words within a single topic, often using measures like pointwise mutual information, cosine similarity, or other metrics between word pairs. Higher coherence scores indicate that the words within a topic strongly relate to each other, suggesting that the topic is well and comprehensible. In other words, higher coherence values typically indicate topics with more semantic coherence, which are easier for humans to interpret and understand. This metric is crucial for determining the optimal number of topics in a model, as it helps identify the configuration.

In the following experiments, the gensim package's coherence score is used to evaluate the coherence of topics produced by topic models LDA within the Python gensim library. The C_V Measure uses a sliding window, a one-set segmentation of the top words, and an indirect confirmation measure that uses normalized pointwise mutual information (NPMI) and the cosine similarity.

3.4 Sentiment Analysis

Sentiment analysis is a method that identifies the emotional state or sentiment behind a situation, often using NLP to analyze text data. Language serves as a mediator for human communication, and each statement carries a sentiment, which can be positive, negative, or neutral[9]. Vader (Valence Aware Dictionary and Sentiment Reasoner)[10] is a rule-based sentiment analysis tool that is specifically designed for analyzing social media texts. Vader is a pre-trained sentiment analysis model that provides a sentiment score for a given text.

Procedures for Sentiment Analysis:

- (1) Basic Python Libraries (import Libraries: pandas, re, nltk...)
- (2) Natural Language Processing (tokenization, normalization, lemmatization, stop words and tf-idf)
- (3) Sentiment Scoring (using Vader)
neg: the negative sentiment score (between 0 and 1)
neu: the neutral sentiment score (between 0 and 1)
pos: the positive sentiment score (between 0 and 1)
compound: the overall sentiment score (between -1 and 1)
- (4) Evaluation.

3.5 Data Visualization

Word clouds can be used to display words with meta-data attached to them. It is a method of seeing how frequently words appear in a particular amount of text by increasing the size of each word proportionally to its frequency. A word cloud is a collection, or cluster, of words of varying sizes. The larger and bolder the term appears, the more frequently it is mentioned in a specific text and the more significant it is. In this text mining project, we combine this wordcloud with Matplotlib, a great tool with a wide range of uses, as well as term frequency and tf-idf approaches, to make it easier to determine which information to focus on in an analysis work.

4 Analysis

In this section, the analytical workflow will be described to enhance comprehension of how the research addresses the three principal questions. Furthermore, each question will be elaborated upon to provide a clearer insight into the execution of the analysis.

4.1 The Flow of Analysis

The analysis section is illustrated in Figure 4.1. The text mining process consists of four main stages, which are Data Preprocessing, Exploration Data Analysis (EDA), Keywords Extraction, and Sentiment Analysis, respectively.

Data Preprocessing is the foundational stage that prepares the raw data for analysis. It includes tokenization to break text into smaller units, stop words removal to filter out common words that add little semantic value, and stemming or lemmatization to reduce words to their base or root form. Finally, the data is combined and sampled to create a manageable and representative dataset. Subsequently, Exploration Data Analysis (EDA) is implemented. The aim of this stage is to delve into the preprocessed data to uncover patterns and insights. This involves analyzing the frequency of words, generating statistical results to understand data distributions and relationships, and employing visualizations to make the data insights accessible and understandable. Keywords Extraction focuses on identifying the most relevant terms within the data. Techniques like TF-IDF (Term Frequency-Inverse Document Frequency) are used to evaluate how important a word is to a document in a collection, and topic modeling, particularly using Latent Dirichlet Allocation (LDA), helps in discovering abstract topics within the document corpus. The final stage is Sentiment Analysis that involves analyzing the sentiments expressed in the texts. This is achieved by selecting a pre-trained model to classify the sentiment as positive, negative, or neutral. The model is then evaluated to assess its performance, and the results are visualized to provide a clear depiction of sentiment trends across the dataset.

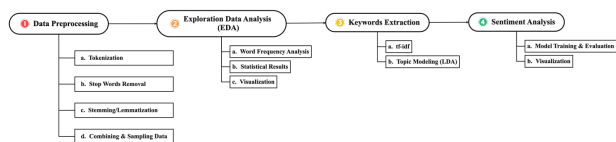


Figure 4.1 The process flow of analysis

4.2. Analysis of Question 1

To address the features of popular books, we start by acquiring a dataset from Kaggle containing reviews for books sold on Amazon. This dataset is refined by applying specific criteria to identify what constitutes a “popular book”. The data is then cleaned to focus on relevant attributes, which are ‘Title’, ‘authors’, ‘publishedDate’, ‘categories’, ‘review/score’, ‘review/summary’, ‘review/text’.

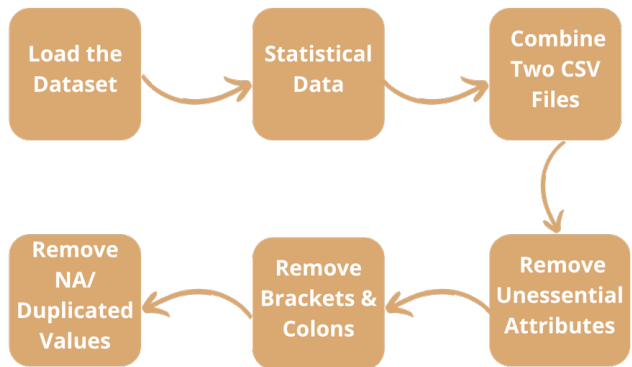


Figure 4.2 The preprocessing flow in question 1

The initial stage in the analytical process involves a detailed data preprocessing workflow, as depicted in the Figure 4.2. This workflow begins with loading the dataset, where the primary data collection is imported for processing. Following this, two CSV files (books_data.csv and Books_rating.csv) containing different segments of the data are combined into a single dataset to unify all relevant information. The next steps focus on cleaning the data, which is crucial for accurate analysis. This includes the removal of NA (not available) and duplicated values to ensure data integrity. Additionally, the data undergoes a cleansing process where brackets and colons are removed to standardize the text format. Finally, unessential attributes are eliminated from the dataset, streamlining the data by discarding redundant or irrelevant information. This meticulous preprocessing not only enhances the quality of the data but also optimizes it for subsequent feature extraction.

The next step is to perform EDA, aiming to discover and ensure the scope of work involved in identifying the features of popular books. EDA serves as a critical phase where various aspects of the dataset are visually and statistically examined to uncover underlying patterns, trends, and anomalies. The EDA process commences with showcasing the differences in the dataset before and after preprocessing. This comparison is vital as it highlights the efficacy of the cleaning and organizing stages, illustrating the improvements in data quality and usability. Following this, the analysis narrows down to specific aspects of the dataset. We explore the top 10 category of books to understand which genres dominate the market. This insight can reveal potential trends in reader preferences and market demands. Additionally, the analysis delves into identifying the books in the fiction category that have the most reviews or texts. This helps in pinpointing what makes these books stand out in terms of reader engagement and popularity. Furthermore, the top 10 authors within the fiction category who have received 5-star ratings are highlighted. This aspect of the analysis sheds light on the authors who consistently produce high-quality content that resonates well with readers, potentially influencing the popularity of their books. Lastly, the number of fiction books published by year is examined. This trend analysis helps in understanding the dynamics of book publication over time, indicating shifts in the

publishing industry's focus towards fiction. Overall, the EDA provides a clear view of the dataset, laying a robust foundation for more detailed analysis concerning the features that contribute to the popularity of books.

The Keywords Extraction stage of the analysis is to derive meaningful insights into the textual data pertaining to popular fiction books, as illustrated in Figure 4. 3.. The process begins by selecting two scopes of data to create corresponding corpora, specifically targeting content related to the most reviewed and popular fiction books. Following the selection, the data undergoes a series of text normalization processes starting with tokenization, where the text is broken down into manageable pieces or tokens. These tokens are then normalized to ensure consistency, for example, by converting all characters to lowercase and correcting common misspellings. Next, stopping words are removed to ensure that the analysis focuses on the most informative words. The cleaned tokens are further refined through stemming and lemmatization, which reduces words to their root forms, thus consolidating similar forms of a word into a single representation. The refined tokens are then analyzed using the Term Frequency-Inverse Document Frequency (TF-IDF) method to identify words that are important to a document but not common across all documents, providing a weighted importance to each term. Following TF-IDF, topic modeling is applied, typically through methods such as Latent Dirichlet Allocation (LDA), to uncover thematic structures in the text. In addition, the Coherence Score Metric is applied to decide the optimal number of topics, thereby providing the best interpretability. All in all, this approach serves two primary goals (two different corpora):

1. Identifying the key features of the top 10 most reviewed books in the fiction category, providing insights into what makes these books resonate with readers.
2. Analyzing the trends in fiction over the recent 10 years, highlighting changes in themes and topics that have captured the interest of readers over time.

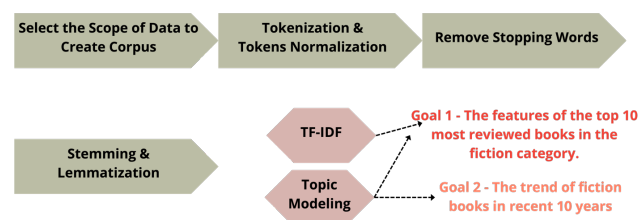


Figure 4.3. The flow of keywords extraction

4.3. Analysis of Question 2

To address the characteristics of Books category, a comprehensive workflow is constructed for analyzing book reviews using the Latent Dirichlet Allocation (LDA) model to identify topics. The process begins with the loading of book review data from two CSV files. This step includes initial data inspection and cleaning, focusing on handling missing values and irrelevant data to ensure quality inputs for analysis. Following data cleaning, the text data undergoes preprocessing to prepare it for LDA analysis. This includes tokenization, where text is split

into meaningful units, and removal of punctuation, numbers, separators, and symbols. The text is also converted to lowercase to maintain consistency. Before delving into topic modeling, EDA is conducted to gain insights into the data's underlying patterns. This includes visualizations of word frequency. The preprocessed text data is then used to train an LDA model. This model helps in discovering the latent topics that are present across the book reviews. Various parameters, such as the number of topics, are assigned directly to find the most meaningful and coherent topics. After the LDA model is applied, the results are evaluated to understand the prevalence and distribution of topics within the data. This includes reviewing the top words in each topic to gauge topic quality and relevance.

4.4. Analysis of Question 3

To understand why the customers complain, sentiment analysis is performed based on question 2. The sentiment analysis model is trained on the preprocessed review data. This section also includes evaluation of the model's performance to ensure it accurately classifies the sentiment of the reviews. Additionally, TF-IDF and wordcloud are also presented to better acquire the results.

5 Results

The results section will present the two csv files' structures to provide dataset background. Further, the experimental results of each question will be shown and interpreted. The question 1 will be executed on local for data preprocessing and EDA and the remaining part will be on google Colab for keywords extraction. As for question 2 and 3, both full programs execute on Kaggle Notebook. The full project utilizes Python as main programming language, nltk, scikit-learn, genism, and pyLDAvis as main packages dealing with NLP.

5.1. Amazon Books Reviews Datasets

The amazon books reviews datasets consist of csv files. One is books_data.csv and the other is Books_rating.csv. The former has 212404 unique books, which provide their related information, such as authors, image, publisher, etc. and the latter incorporates 3M user on 212404 unique books the dataset that contains product reviews and metadata from Amazon, including 142.8 million reviews spanning May 1996 to July 2014. The more detail datasets information is shown at Table 5.1 and 5.2.

Table 5.1 The information of books_data.csv

RangeIndex: 212404 entries, 0 to 212403				
Data columns (total 10 columns):				
#	Column	Non-Null	Count	Dtype
0	Title	212403	non-null	object
1	description	143962	non-null	object
2	authors	180991	non-null	object
3	image	160329	non-null	object
4	previewLink	188568	non-null	object
5	publisher	136518	non-null	object
6	publishedDate	187099	non-null	object
7	infoLink	188568	non-null	object
8	categories	171205	non-null	object
9	ratingsCount	49752	non-null	float64
dtypes: float64(1), object(9)				

Table 5.2. The information of Books_rating.csv

RangeIndex: 3000000 entries, 0 to 2999999		
Data columns (total 10 columns):		
#	Column	Dtype
0	Id	object
1	Title	object
2	Price	float64
3	User_id	object
4	profileName	object
5	review/helpfulness	object
6	review/score	float64
7	review/time	int64
8	review/summary	object
9	review/text	object
dtypes: float64(2), int64(1), object(7)		

5.2. The Results of Question 1

For the data preprocessing steps as previously mentioned, the two datasets will be combined based on Title attribute, further only specific attributes are retained for next step, which shown in Table 5.3. To remove brackets and colons, regular expression is used `str.extract(r'\(.*\)')`. In addition, `lambda x: str(x)[:4]` function is applied to unify the publishedDate attribute by keeping the year only. By using `drop.na()` and `drop_duplicates()` functions, NA and duplicated values can be easily removed. Table 5.4 shows the detail after data preprocessing. It is clear that the numbers of records are significantly reduced, thereby ensuring further explorations' integrity and enhance the computation speed.

Table 5.3 The dataset after combining and attributes selection

RangeIndex: 3000000 entries, 0 to 2999999		
Data columns (total 7 columns):		
#	Column	Dtype
0	Title	object
1	authors	object
2	publishedDate	object
3	categories	object
4	review/score	float64
5	review/summary	object
6	review/text	object
dtypes: float64(1), object(6)		

Table 5.4 The dataset after data preprocessing

Index: 2091935 entries, 0 to 2999995		
Data columns (total 7 columns):		
#	Column	Dtype
0	Title	object
1	authors	object
2	publishedDate	object
3	categories	object
4	review/score	float64
5	review/summary	object
6	review/text	object
dtypes: float64(1), object(6)		

The EDA explores several interesting phenomena related to the question. Figure 5.1 presents the top 10 category that inside the processed dataset. The fiction type is considerably higher than other books type. Further, we calculate which fiction books have the most review text, and the result is shown at Figure 5.2, we could know that Harry Potter & the Hobbit are the top 3 fiction books with the most review text. We also discover the top 10 Authors with 5 star Ratings in Fiction Category, J. R. R. Tolkien, J. K. Rowling, and Rey Trciero are among top 3, as depicted in Figure 5.3. Furthermore, the number of fiction books published by year is presented in Figure 5.4, the time range is from 1931 to 2022 and the peak of publishing fiction is around 2010.

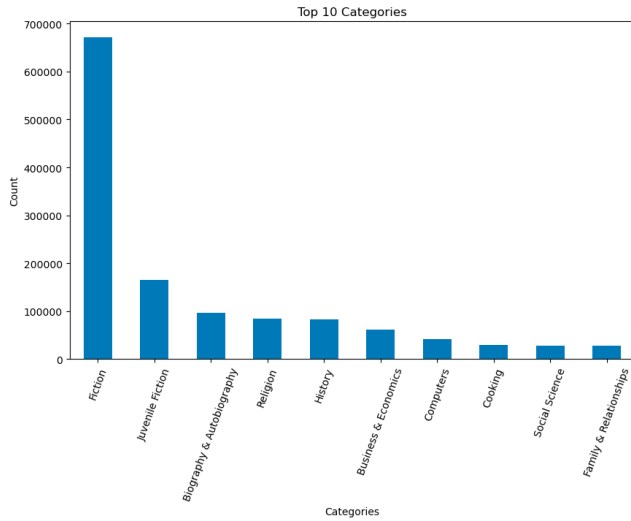


Figure 5.1 The top 10 category

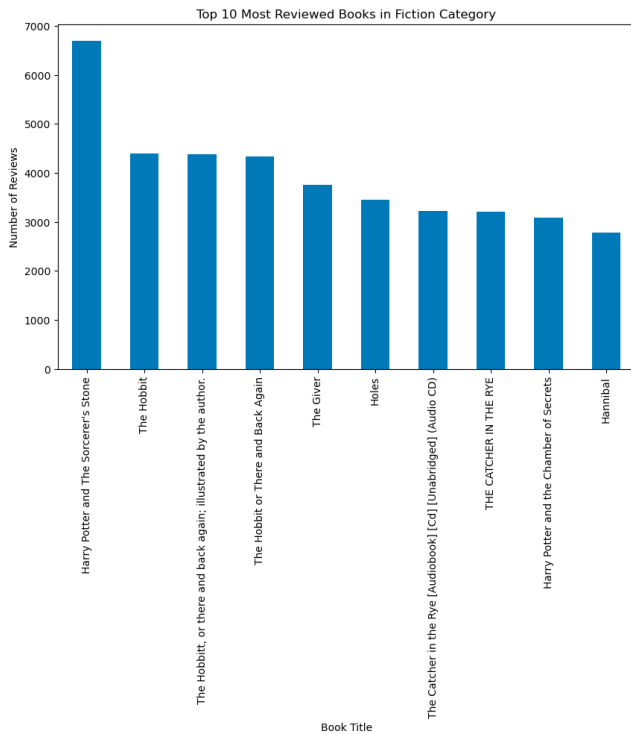


Figure 5.2 The top 10 most reviewed books in fiction category

For text mining, the first step is to select the scope of data to create corpus. Given that we have two primary goals, there are 2 corpora will be constructed. The criteria of the first are the following,

1. Identify the top 10 most reviewed books in the fiction category.
2. Filter reviews that have more than 100 words.

3. Filter books that were published from the year 1900 onwards.

This selective approach ensures that the corpus is both relevant and manageable, focusing specifically on substantive reviews of popular fiction books from a significant period in literary history. This targeted dataset will facilitate a more focused analysis of the textual elements that resonate with readers and contribute to a book's popularity. The wordcloud (Figure 5.6) focus on reviews of popular fiction books. The notable terms like "Hobbit," "Holden," and "Potter" suggest that specific works or characters such as "The Hobbit," "The Catcher in the Rye" (Holden Caulfield), and "Harry Potter" are heavily discussed, indicating their popularity and influence within the corpus. And words such as "great," "good," "like," and "love" highlight positive sentiments expressed in the reviews. These adjectives provide insight into the general reception of the books within this corpus. The presence of words like "character," "life," "world," and "story" points to discussions centered around character development, world-building, and plotlines, which are critical elements in fiction that resonate with readers.

The analysis for determining the optimal number of topics for our LDA model involves plotting coherence scores against varying topic counts, as shown in Figure 5.6. The graph indicates a sharp decline in coherence as the number of topics increases from 5 to 10, followed by fluctuations that suggest less stability in the topic quality with higher numbers of topics. Based on this analysis, a model with 5 topics was selected as it provided the highest coherence score. The LDA model with these parameters (5 topics, 15 passes through the corpus, and 100 iterations), the topics reflected distinct thematic concentrations within the corpus:

1. The first topic highlights general literary elements, emphasizing terms like "character," "write," and "well," suggesting discussions around narrative structure and character development.
2. The second topic is clearly dominated by references to "Harry Potter," with associated words like "school" and "first," pointing to key aspects of the Harry Potter series.
3. The third topic focuses on "The Hobbit" with terms like "Bilbo," "Stanley," and "camp," indicating a concentration on adventure and journey narratives.
4. The fourth topic includes words like "Jonas," "world," "giver," "life," and "memory," which are central to the themes of "The Giver," a novel that explores dystopian realities and philosophical reflections on society and memory.
5. The fifth topic centers around "Holden," the protagonist of "The Catcher in the Rye," with associated terms reflecting the introspective and critical nature of the narrative, such as "life," "think," and "school."

The topics identified encompass a certain range of literary discussions, from character development to thematic explorations of specific novels, which may suggest that the specific novels have superior impact in the constructed corpus.

The criteria of the second corpora are the following,

1. Recent 10 years
2. The review/score should be at least 4
3. The word count should be at least 100

The coherence scores, as depicted in Figure 5.7, suggest that the number of topics was optimized for clarity and depth, with a peak coherence at 12 topics. The 12 topics are listed as following,

- These topics reveal themes prevalent in contemporary fiction,

Topics like life, humanity, and the world reflect universal existential concerns (Topic 0).

Specific topics highlight intensive discussions on classic literature such as Melville's "Moby Dick" (Topic 1) and Hemingway's works (Topic 2).

Topics focusing on characters and relationships, like Jane Austen's explorations of love and marriage (Topic 4), and broader social issues encompassing family and personal growth (Topic 6).

Distinct genres are identified, such as war (Topic 7), children's literature (Topic 8), and religious texts (Topic 10), each representing a focused area of reader and critical interest.

The love for mystery and adventure is captured in topics discussing mysteries, murders, and ongoing series (Topic 11).

Authors	Number of 5-Star Reviews
J.R.R. Tolkien	13000
J.K. Rowling	8200
Ray Bradbury	5000
John Steinbeck	4800
J.R. Zobol	4500
Kurt Vonnegut	4400
Stephen King	4300
Jane Austen	4200
George Orwell	4200
Harper Lee	3500

The scatter plot, titled "Number of Fiction Books Published by Year", displays the annual count of fiction books published from approximately 1700 to 2010. The y-axis, labeled "Number of Books Published", ranges from 0 to 60,000 in increments of 10,000. The x-axis, labeled "Year", ranges from 1700 to 2010, with major ticks every 50 years. The data shows a long period of low publication volume (mostly below 10,000 books per year) until the mid-20th century. Starting around 1950, there is a rapid and significant increase in the number of books published, reaching a peak of over 60,000 books around 2002. Following this peak, the volume shows a sharp decline, with a notable gap in data around 2009, and then a slight recovery in the early 2010s.

Figure 5.5 The wordcloud of corpus 1

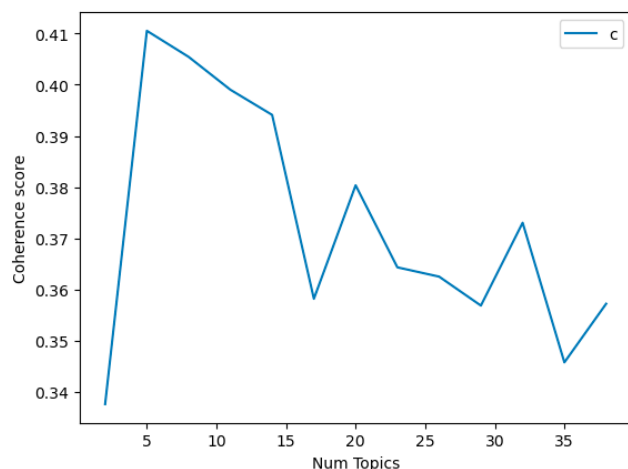


Figure 5.6 The coherence score of corpus 1

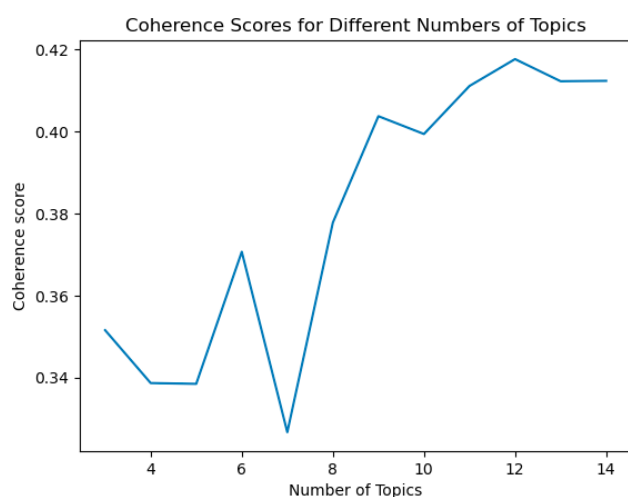


Figure 5.7 The coherence score of corpus 2

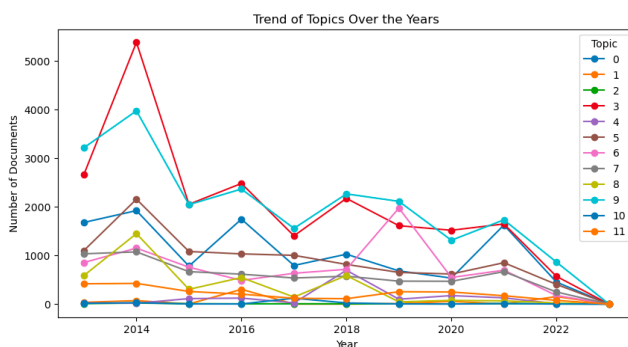


Figure 5.8 The trend of topics in corpus 2

5.3. The Results of Question 2

The question 2 focus on the characteristics of Books category, focusing on the book “The Death of Outrage: Bill Clinton and the Assault on American Ideals”. The LDA results, as shown in Figure 5.9, show five distinct topics that reveal the breadth of themes covered in the dataset.

1. Topic 0
This topic likely represents books that focus on American political and social issues, discussing themes like patriotism, governance, public sentiment, and moral questions within the U.S. context.
2. Topic 1
The frequent mention of “president,” “political,” “Reagan,” and “lie” suggests this topic includes political biographies or critical analyses focusing on political figures and ideologies, with a potential emphasis on controversies and leadership.
3. Topic 2
This topic appears to delve into books that discuss ethical and moral arguments, touching on morality, political behavior, and the implications of leadership decisions on the public and the nation.
4. Topic 3
Featuring a mix of words like “write,” “moral,” “right,” and “William,” this topic may cover historical and cultural analysis, possibly including biographies, essays, and scholarly works on American history or specific influential figures.
5. Topic 4
This topic seems to focus on contemporary social issues, evidenced by terms like “life,” “news,” “truth,” “sex,” and “important,” suggesting a focus on modern social challenges, media influence, and public discourse.

	Topic 0	Topic 1	Topic 2	Topic 3	Topic 4
0	american	outrage	president	president	outrage
1	outrage	president	moral	american	moral
2	make	moral	argument	write	president
3	people	people	make	moral	could
4	country	lie	american	right	write
5	president	american	outrage	america	life
6	think	death	morality	outrage	million
7	moral	point	people	morality	course
8	take	think	america	william	much
9	get	make	show	get	call
10	much	review	public	state	good
11	great	public	behavior	time	take
12	today	know	get	argument	action
13	give	say	political	make	lack
14	never	never	death	go	news
15	say	find	say	fact	bennet
16	come	reader	need	real	truth
17	defend	right	come	need	sex
18	understand	political	country	show	fact
19	point	reagan	good	people	important

Figure 5.9 The LDA results of question 2

Overall, the LDA results illustrate that the Books category encompasses a diverse range of topics primarily centered around political, ethical, cultural, and social themes. The words they

contain are often related to themes such as Politics, Biography, or Non-fiction.

5.4. The Results of Question 3

The question 3 further delves into the sentiment of customers in Books category, concentrating on the specified book.

The distribution of review scores shows a heavy concentration of high scores, depicted in Figure 5.10, with most reviewers rating the book either 1 or 5, indicating polarized opinions about the content. The large number of 1-star and 5-star reviews suggests that while some readers strongly agree with the author’s perspective, others are extremely opposed or dissatisfied.

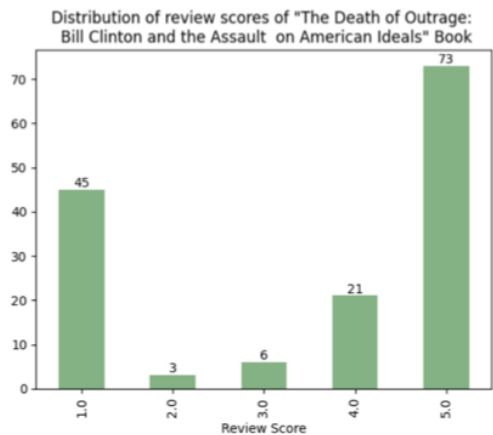


Figure 5.10 The distribution of review scores of a specific book

And the following wordcloud (Figure 5.11) shows prominently features words like “moral,” “lie,” “president,” “republican,” and “outrage,” suggests that much of the discussion and likely complaints center around political and moral arguments.

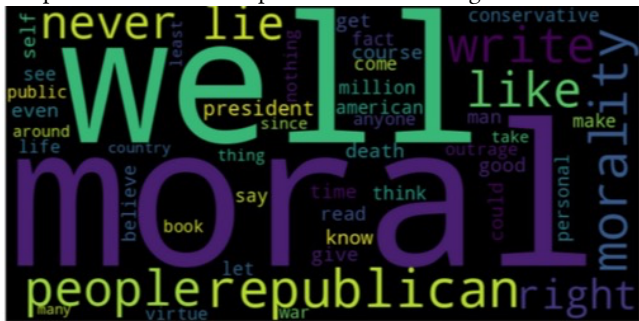


Figure 5.11 The wordcloud regarding complaining

The compound score is predominantly negative that below - 0.15 (Figure 5.12), indicating that even among those who rated the book highly, there is a sentiment of negativity or criticism in the text of the reviews. The sentiment breakdown (Figure 5.13) shows that most of the sentiment is neutral, followed by a smaller proportion of negative sentiment and very little positive sentiment. This could indicate that while the reviews are critical,

they might be more focused on content discussion rather than purely emotional reactions.

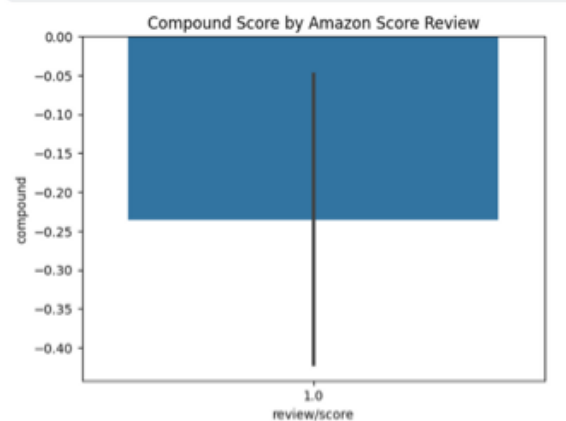


Figure 5.12 The compound score of a specific book

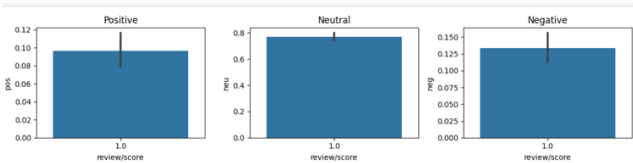


Figure 5.13 The breakdown sentiment score of a specific book

6 Limitations and Conclusion

The results of our Exploratory Data Analysis (EDA) in question 1 provide a foundational understanding of the characteristics that popular books and highly rated authors share. By using (unsupervised) text mining techniques such as TF-IDF and LDA, we can extract key words/themes and stylistic features that correlated with readers. These insights can identify what makes books popular and offer guidance for publishers and authors aiming to meet reader expectations.

The biggest limitation of question 1 is the data scope and selection bias because the dataset might only include books from certain publishers, genres, or authors, particularly if it is sourced from a specific platform. The second is temporal changes. The habit of readers can change over time, and features that are prevalent in popular books during one period may not be as relevant in another. If the data does not account for temporal variations or is not regularly updated, it may not accurately reflect current trends. In addition, the published books may be read after several years later, the analysis of trend may not be accurate and precise. Furthermore, the interpretability is highly subjective in topic modeling, this may not fully express the essence of popular books.

Specifically, in Question 2, we successfully classified the book “The Death of Outrage: Bill Clinton and the Assault on American Ideals” into the potential genres of “Politics,” “Biography,” and “Non-fiction.” This achievement offers a promising direction for

future work, where we aim to categorize all books under the label "Books." However, we were unable to achieve our goal with VADER when analyzing books rated with a score of 1 in Question 3, we need find another way to deal with "sarcasm", "negation" or "multipolarity" which also are not quite trouble-free even for human in conversation to filter. Moving forward, we will expand our analysis for Questions 2 and 3 by looking into more advanced methodologies and models. This will allow us to better manage enormous datasets and address the challenges of sentiment analysis.

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